

Tidal Energy Harvesting via Floating Reservoir Systems: A Proposed Framework for Global Renewable Energy, Freshwater Distribution, and Superconducting Transmission

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Abstract

This paper proposes a comprehensive framework for harvesting tidal energy through networks of large floating reservoir systems deployed across oceans and seas. By capturing water at high tide and directing it through turbines, the system can generate continuous electricity without combustion or radioactive materials. The generated power is proposed to support global freshwater desalination and distribution — including to arid regions such as the Sahara Desert — and to be transmitted over long distances via superconducting power lines. Design considerations for pool geometry, structural stability, ecosystem impact mitigation, and transmission infrastructure are discussed.

1. Introduction

The global demand for clean, renewable energy continues to grow, yet existing solutions such as solar, wind, and conventional hydroelectric power face significant geographic and intermittency constraints. Tidal forces — driven by the gravitational interaction between the Earth, Moon, and Sun — represent a largely untapped, highly predictable energy source available across the world's coastlines and open oceans. This paper outlines a proposed engineering concept for exploiting tidal energy at scale through the deployment of floating pool-based reservoirs, coupled with a superconducting transmission grid and a desalination infrastructure to address both energy and freshwater scarcity.

2. System Design and Operating Principle

The core concept involves the construction of large floating reservoirs (pools) moored to the seabed by multiple tethering ropes. Each pool is designed to fill passively with seawater near the moment of peak high tide. The stored water is subsequently directed through turbines positioned around the perimeter of the reservoir to generate electricity. To enable continuous generation — including during the high-tide period itself — a secondary elevated reservoir is proposed. A portion of the captured water would be pumped to this upper-level pool using electricity generated by the system; the resulting hydraulic head would then drive turbines without interruption throughout the tidal cycle. Water discharged from the upper reservoir can be collected in the lower pool and recirculated, improving overall system efficiency.

3. Structural Stability and Pool Configuration

Circular pool geometries are recommended for several reasons. Circular forms distribute hydraulic and mechanical stress more uniformly than rectangular alternatives, and are also more easily recognized by marine wildlife (particularly cetaceans), potentially reducing the risk of collision. Horizontal stabilization is achieved through a distributed mooring system in which each section of the pool floor is independently connected to the seabed via a tether-and-airbag assembly. The airbags, positioned at appropriate depth, maintain vertical buoyancy while metal stabilizer bars prevent horizontal drift. Individual pool segments are joined by impermeable flexible membranes, allowing the structure to accommodate ocean swells and current-driven movement. Adjustable wall heights allow operators to regulate water intake volume based on predicted tidal amplitude.

4. Environmental Considerations

Minimizing ecological disruption is a central design requirement. Pool floors are proposed to be constructed from semi-transparent or transparent materials to allow solar radiation to penetrate to sub-surface depths, thereby reducing interference with photosynthetic marine organisms. The calm surface conditions created by a dense array of floating pools may also provide incidental benefits during tsunami events, as energy from the wave front would

be partially absorbed by the filling of surface reservoirs. Careful hydrodynamic modeling of local current patterns and marine habitat surveys would be required prior to deployment in any given region.

5. Power Transmission Infrastructure

Conventional high-voltage transmission lines are limited in their effective range to approximately 400–500 km due to resistive losses. To connect offshore tidal arrays to inland consumption centers and to enable intercontinental energy distribution, the deployment of superconducting transmission cables is proposed. Offshore superconducting segments would be maintained near absolute zero using cryogenic cooling systems; flexible insulation would be applied at the points where conductors exit their cooling housings to ensure both electrical safety and mechanical adaptability to ocean movement. Elevated transmission towers connecting pool arrays would be engineered with sufficient clearance for the passage of large ocean-going vessels.

6. Freshwater Production and Distribution

The electricity generated by tidal pool networks is proposed to power large-scale desalination plants capable of converting seawater and brackish water into potable water. The distribution of this water via pipeline and pumping infrastructure — itself powered by the same tidal grid — could extend reliable freshwater access to currently underserved regions, including the Sahara Desert and other arid zones. This integrated approach could simultaneously address global energy insecurity and water scarcity within a single infrastructure framework.

7. Conclusion

The proposed tidal floating reservoir system offers a conceptually novel approach to renewable energy generation that exploits a globally available, fully predictable natural phenomenon. By combining tidal energy capture with superconducting transmission technology and desalination infrastructure, the framework has the potential to address two of humanity's most pressing resource challenges at scale. Further research is required to evaluate structural engineering feasibility, marine ecological impact, cryogenic system integration in open-ocean environments, and the economic viability of large-scale deployment.

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